

DAVIDE ITALO SERRAMAZZA

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Professional Summary

PhD in Explainable AI (XAI) for Time Series with 7+ first-author publications and a software library (tsCaptum) on PyPI. Proven ability to bridge academic research and production-ready code. Seeking research scientist or ML engineer roles requiring deep expertise in model interpretability, model deployments and time series analysis.

Education

PhD in XAI for Time Series

University College Dublin (UCD)

September 2022 – September 2026

Dublin, Ireland

Master in AI (School of Computer Science), Grade 110/110

Università di Pisa

September 2017 – March 2020

Pisa, Italy

Bachelor in Computer Science

Università di Pisa

September 2013 – September 2017

Pisa, Italy

Experience/Projects

University College Dublin

PhD in AI for Time Series

September 2022 – September 2026 (thesis submission)

Dublin, Ireland

- Adapted XAI method as SHAP, Lime, Feature Ablation for Time Series
- Developed **tsCaptum**, a Python library bridging XAI library (Captum) with Time Series library (aeon, sktime). Computational time reduced by 1-2 orders of magnitude, performance not hurt.
pypi.org/project/tsCaptum/
- Published 4 papers as first author scholar.google.com/citations?user=DFHxPqMAAAAJ&hl
- Technical stack:* Python, PyTorch, Captun, aeon, sktime, scikit-learn, GitHub

Università di Pisa / Mediateca Toscana

Assegno di Ricerca (Research job)

May 2020 – May 2022

Tuscany, Italy

- Built an **API + database** to serve video clips to a web platform
- Processed 100+ hours of documentary clips using deep learning algorithms
- Technical stack:* Python, PyTorch, TensorFlow, GitHub, Node.js, MongoDB, REST APIs

Technical Skills

- Programming Languages:** Python, Nodejs, Matlab, Java, JavaScript, C++, C, Bash script
- AI Frameworks:** Pytorch, TensorFlow, Sktime, Aeon, Scikit-learn, Captum
- XAI:** SHAP, KernelSHAP, LIME, Feature Ablation, dCAM)
- Technologies/Frameworks:** Linux, GitHub, Docker, CUDA acceleration

Languages

- Italian** (native)
- English** (Professional working proficiency)